PROBLEM 1 | Problems T2_CrystStruct.pdf | Crystalline Structures & Miller Indices

Miller Indices and Densities in FCC

Original statement · Problems T2_CrystStruct.pdf

1. Find the Miller indices corresponding to the planes in the figures: Figure I, Figure II, Figure III. For an FCC structure, with lattice parameter a, calculate the linear density along direction $[110]$, the planar density on the plane drawn on figure I and the volumetric density. *(Solution: I $(\bar{1}1\bar{1})$; II $(\bar{2}31)$; III $(\bar{1}\bar{1}1)$; $\rho_l = \sqrt{2}/a$; $\rho_s = 4/(a^2\sqrt{3})$; $\rho_v = 4/a^3$)*

Solution: **I** $(\bar{1}1\bar{1})$; **II** $(\bar{2}31)$; **III** $(\bar{1}\bar{1}1)$; $\rho_l = \sqrt{2}/a$; $\rho_s = 4/(a^2\sqrt{3})$; $\rho_v = 4/a^3$

PHASE 1: HYPOTHESIS & PARAMETERS

Physical Context:

Three crystallographic planes drawn inside cubic cells of edge a are analysed. Afterwards, a Face-Centred Cubic (FCC) lattice of lattice parameter a is evaluated.

Input Data:

- **Figure I:** The plane passes through the upper right vertex, intersecting the axes at: $x = -a$, $y = a$, $z = -a$ (taking a suitable origin).
- **Figure II:** The plane cuts the axes at $x = -a/2$, $y = a/3$, $z = a$.
- **Figure III:** The plane cuts the axes at $x = -a$, $y = -a$, $z = a$.
- **FCC structure:** Lattice parameter a, number of atoms per cell $n = 4$, atomic radius $R = a\sqrt{2}/4$.

Starting Hypotheses:

1. The cubic lattice is orthonormal ($a = b = c$, $\alpha = \beta = \gamma = 90^\circ$).
2. For planes with negative indices, the coordinate origin is conveniently translated to an adjacent vertex of the unit cell [Session 3 Slide 49].
3. Atoms are modelled as rigid spheres in contact along the face diagonals $\langle 110 \rangle$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Before calculating, we identify the governing principles of the official syllabus:

- **Determination of Miller Indices** $(h\,k\,l)$ **[Session 3 Slides 44–45]:**
- Identify the fractional intercept coordinates (p, q, r) with the axes x, y, z.
- Compute the reciprocals $(1/p, 1/q, 1/r)$.
- Reduce to the smallest set of integers by multiplying by the LCM.
- **Linear Density** (ρ_l) **[Session 4 Slide 8]:**

$$\rho_l = \frac{N_{\text{atoms centred on the segment}}}{L_{[uvw]}}$$

- **Planar Density** $(\rho_s$ **or** $\rho_p)$ **[Session 4 Slide 9]:**

$$\rho_s = \frac{N_{\text{atoms contained in the plane}}}{A_{(hkl)}}$$

- **Volumetric Density** (ρ_v) **[Session 4 Slide 7]:**

$$\rho_v = \frac{n}{V_C} = \frac{n}{a^3}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Miller Indices of the Planes:

- **Figure I:**
- Translating the origin to the corner $(1, 0, 1)$, the plane intersects the axes at:

$$x = -1, \quad y = +1, \quad z = -1$$

- Reciprocals of the intercepts:

$$\frac{1}{-1} = -1, \quad \frac{1}{+1} = 1, \quad \frac{1}{-1} = -1$$

- Miller notation:

$$(\bar{\mathbf{1}}\mathbf{1}\bar{\mathbf{1}})$$

(Note that it belongs to the close-packed family of planes $\{111\}$*).*

- **Figure II:**
- Reading the marks on the axes of Figure II:
- x axis: intercept at $-a/2 \implies p = -1/2$ (with origin translated to $x = 1$).
- y axis: intercept at $+a/3 \implies q = +1/3$.
- z axis: intercept at $+a \implies r = +1$.
- Reciprocals:

$$\frac{1}{-1/2} = -2, \quad \frac{1}{1/3} = 3, \quad \frac{1}{1} = 1$$

- Miller notation:

$$(\bar{\mathbf{2}}\mathbf{3}\mathbf{1})$$

- **Figure III:**
- Translating the origin to $(1, 1, 0)$:
- x axis: intercept at $-1 \implies p = -1$.
- y axis: intercept at $-1 \implies q = -1$.
- z axis: intercept at $+1 \implies r = +1$.
- Reciprocals:

$$\frac{1}{-1} = -1, \quad \frac{1}{-1} = -1, \quad \frac{1}{1} = 1$$

- Miller notation:

$$(\bar{\mathbf{1}}\bar{\mathbf{1}}\mathbf{1})$$

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2. Calculations for the FCC Lattice:

- **Linear density along** $[110]$**:**
- The direction $[110]$ is the diagonal of the bottom face of the cube.
- Segment length: $L_{[110]} = a\sqrt{2}$.
- Number of atoms cut diametrally:

$$N_{\text{atoms}} = 2 \times \left(\frac{1}{2}\right)_{\text{vertices}} + 1_{\text{face centre}} = 1 + 1 = 2 \text{ atoms}$$

- Linear density:

$$\rho_l = \frac{2}{a\sqrt{2}} = \frac{2\sqrt{2}}{2a} = \boxed{\frac{\sqrt{2}}{a}}$$

- **Planar density on the plane of Figure I** $(\bar{1}1\bar{1}) \in \{111\}$**:**
- The $\{111\}$ plane intersects three face diagonals of length $L = a\sqrt{2}$, forming an equilateral triangle.
- Area of the triangle:

$$A = \frac{\sqrt{3}}{4}L^2 = \frac{\sqrt{3}}{4}(a\sqrt{2})^2 = \frac{\sqrt{3}}{4}(2a^2) = \frac{\sqrt{3}}{2}a^2$$

- Atoms contained inside the triangle of the plane:

$$N_{\text{atoms}} = 3 \times \left(\frac{1}{6}\right)_{\text{vertices}} + 3 \times \left(\frac{1}{2}\right)_{\text{edges}} = \frac{1}{2} + \frac{3}{2} = 2 \text{ atoms}$$

- Planar density:

$$\rho_s = \frac{2}{\frac{\sqrt{3}}{2}a^2} = \boxed{\frac{4}{a^2\sqrt{3}}}$$

- **Volumetric density (**ρ_v**):**
- The FCC cell has $n = 4$ equivalent atoms in a volume $V_C = a^3$:

$$\rho_v = \frac{\mathbf{4}}{\mathbf{a^3}}$$

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

1. **Dimensionality:**
2. $[\rho_l] = \text{length}^{-1} \implies \frac{\sqrt{2}}{a}$ has dimensions of atoms/m.
3. $[\rho_s] = \text{length}^{-2} \implies \frac{4}{a^2\sqrt{3}}$ has dimensions of atoms/m^2.
4. $[\rho_v] = \text{length}^{-3} \implies \frac{4}{a^3}$ has dimensions of atoms/m^3.
5. **Packing consistency:**

The $[110]$ direction is the line of maximum linear density in FCC ($\rho_l = 1/(2R)$), and the $\{111\}$ plane is the plane of maximum planar density ($\text{APF}_{\text{planar}} = \pi/(2\sqrt{3}) \approx 90.7\%$), which shows why $\{111\}\langle 110\rangle$ is the canonical slip system of FCC metals.